



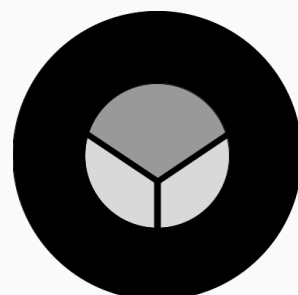
Down the River

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DE Intro to Statistics I

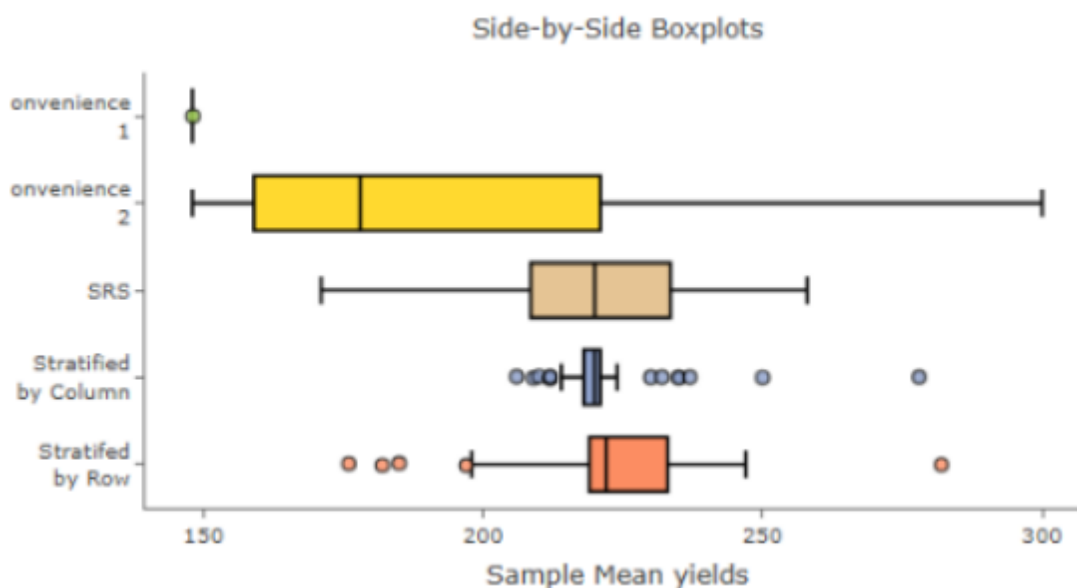
September 24, 2025

“Truth is ever to be found in the simplicity, and not in the multiplicity and confusion of things” - Issac Newton



Instructions: Use the graphical displays and statistical summaries to answer the questions. Recall, the farmer is going to use the results of a sample ($n = 10$) to determine whether it is worthwhile to harvest all plots ($N = 100$).

Mean yields	n	Mean	Std. Dev.	Min	Q1	Median	Q3	Max	IQR
convenience 1	137	148.00073	0.00854	148.00000	148.00000	148.00000	148.00000	148.10000	0.00000
convenience 2	137	195.95620	43.15514	148.00000	159.00000	178.00000	221.00000	300.00000	62.00000
SRS	137	220.68613	17.28981	171.00000	208.50000	220.00000	233.50000	258.00000	25.00000
Stratified by Column	137	220.59854	6.79150	206.00000	218.00000	220.00000	221.00000	278.00000	3.00000
Stratified by Row	136	223.82353	13.31437	176.00000	219.00000	222.00000	233.00000	282.00000	14.00000



The farmer could try to estimate the true population mean yield, μ , based on his one sample.

For teaching + learning purposes, and for our analysis, we need to know the population mean, which is $\mu = 220.95$ units. The population standard deviation is $\sigma_y = 51.7$ units.

0.0.1 Why do we sample? Oftentimes the purpose is to estimate unknown/unknowable population parameters like the population mean. With that in mind, order the Sample Designs from best to worst in terms of which design tended to better estimate the actual population mean corn yield.

Since the population mean looks to be between 200 and 250, three of the five distributions fit this description. The SRS is roughly unimodal, with the greatest (out of the three tables with a mean ~ 225) inter-quartile range. The best out of the three in terms of the IQR is the "Stratified by Column", and the second best seems to be "Stratified by Row." This is because the column has a significantly lower IQR than row. Convenience 1 & 2 were inherently flawed in their sampling methods, so a mean of 150 (convenience 1), and a mean of 175 (convenience 2) are not correct in terms of results. In this case, the order will be (best to worst): {Stratified by Column, Stratified by Row, SRS, Convenience 2, Convenience 1}

0.0.2 In 2-4 sentences, make a persuasive statement about why the two convenience sample designs do not provide effective estimators for the actual mean yield of the population ($N = 100$) of plots.

Both convenience 1 & 2 were flawed in their sampling ways. Because they were picked based on patterns that are "easy" for the farmer to achieve, they might actually introduce bias. If the convenience patterns are predicted based on non-random sampling, this will lead to the data not being representative of the population because it introduces human bias.

0.0.3 Use the summary statistics to make a compelling case for why one design is better than the other two when comparing the SRS, Stratified by Column, and Stratified by Row sample designs.

In all honesty, stratifying homogenous groups, by row, is the best case out of the three presented.